

Module 4: Sensor and Measurements

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1 Multicopter Sensors and Measurement Models

Autonomous multicopter flight requires accurate knowledge of attitude, angular velocity, position, velocity, altitude, and heading. However, these quantities are not directly measurable. Instead, they are inferred from onboard sensors through modeling and state estimation.

In the system architecture of Beard and McLain (Chapter 7, Sensors for MAVs), the onboard sensors provide measurements to a state estimator, which in turn provides state estimates to the autopilot and higher-level guidance layers. Thus, control performance is fundamentally limited by sensing and estimation accuracy.

In this section, we examine the primary sensors used in multicopters and derive their measurement models.

2 Accelerometers

2.1 Physical Interpretation

An accelerometer does *not* measure inertial acceleration directly. Instead, it measures *specific force*, defined as

$$f = a - g, \quad (1)$$

where a is the inertial acceleration of the vehicle and g is gravitational acceleration.

Expressed in the body frame,

$$f^b = R_b^i(a^i - g^i), \quad (2)$$

where R_b^i is the rotation matrix from inertial to body coordinates.

Thus, an accelerometer measures Specific force expressed in the body frame.

2.2 Hover Interpretation

At hover,

$$a^i = 0, \quad (3)$$

so that

$$f^b = -R_b^i g^i. \quad (4)$$

Therefore, accelerometers provide information about the vehicle's orientation relative to gravity. This is why accelerometers are useful for roll and pitch estimation.

However, accelerometers cannot distinguish between:

- Horizontal acceleration, and
- Tilt relative to gravity.

This phenomenon is known as the **tilt–acceleration ambiguity**.

2.3 Measurement Model

A typical accelerometer model is

$$\tilde{f} = f + b_a + \eta_a, \quad (5)$$

where

- b_a is a slowly varying bias,
- η_a is measurement noise.

Bias drift must be estimated and compensated in practice.

3 Rate Gyros

3.1 Physical Interpretation

Rate gyros measure the angular velocity of the body frame relative to the inertial frame, expressed in body coordinates:

$$\omega^b = \begin{bmatrix} p \\ q \\ r \end{bmatrix}. \quad (6)$$

These measurements are essential for propagating attitude kinematics.

3.2 Role in Attitude Propagation

The rotational kinematics satisfy

$$\dot{R} = RS(\omega), \quad (7)$$

or, using Euler angles q ,

$$\dot{q} = T(q)\omega. \quad (8)$$

Thus, gyros provide the differential equation input needed to compute attitude through integration.

3.3 Measurement Model

A typical gyro model is

$$\tilde{\omega} = \omega + b_g + \eta_g, \quad (9)$$

where b_g is gyro bias and η_g is noise.

Because attitude is obtained by integration,

$$\theta(t) \sim \int b_g dt, \quad (10)$$

gyro bias leads to unbounded attitude drift if not corrected.

4 Pressure Sensors

Two pressure sensors are commonly used:

1. Static pressure sensor (altitude),
2. Differential pressure sensor (airspeed).

4.1 Static Pressure and Altitude

Atmospheric pressure decreases with altitude. Using the barometric relation, altitude can be approximated as

$$h = \frac{T_0}{L} \left[\left(\frac{P}{P_0} \right)^{-RL/g} - 1 \right], \quad (11)$$

where P is static pressure.

In practice, this relation is linearized for small altitude variations.

Thus, static pressure provides

$$\boxed{\text{Altitude estimate.}}$$

4.2 Differential Pressure and Airspeed

Using Bernoulli's equation,

$$\Delta P = \frac{1}{2} \rho V_a^2, \quad (12)$$

so that

$$V_a = \sqrt{\frac{2\Delta P}{\rho}}. \quad (13)$$

This provides the magnitude of the airspeed vector.

Note that airspeed differs from ground speed in the presence of wind.

5 Magnetometer (Digital Compass)

A magnetometer measures the Earth's magnetic field vector expressed in the body frame:

$$m^b = R_b^i m^i. \quad (14)$$

This measurement is primarily used to estimate heading (yaw angle).

5.1 Practical Issues

Magnetometers are sensitive to:

- Hard iron effects,
- Soft iron distortions,
- Local magnetic disturbances,
- Electrical interference.

Thus, calibration and bias estimation are essential.

6 GPS

The Global Positioning System provides:

- Position in inertial coordinates,
- Ground speed,
- Course angle.

Measurement models:

$$\tilde{p}^i = p^i + \eta_p, \quad (15)$$

$$\tilde{V}^i = V^i + \eta_V. \quad (16)$$

GPS has:

- Low update rate (5–20 Hz),
- High long-term accuracy,
- No direct attitude information.

GPS corrects long-term drift from IMU integration.

7 Time-of-Flight Sensors

Time-of-flight (ToF) sensors measure range by emitting a signal and measuring return time:

$$d = \frac{ct}{2}, \quad (17)$$

where c is wave propagation speed.

These sensors are commonly used for:

- Low-altitude hold,
- Indoor flight,
- Obstacle detection.

8 LIDAR

LIDAR (Light Detection and Ranging) measures distance using laser pulses.

Measurement model:

$$\tilde{d} = d + \eta_d. \quad (18)$$

Applications:

- Obstacle avoidance,
- SLAM,
- Terrain mapping,
- Precision landing.

LIDAR provides high-resolution geometric information but increases power and computational load.

9 Visual Sensors

Cameras provide 2D pixel measurements.

Using projective geometry,

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = K[R \mid t] \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}. \quad (19)$$

Applications include:

- Visual odometry,
- Feature tracking,
- Target localization,
- Optical flow,
- Vision-based landing.

Vision sensors provide rich information but require substantial computation.

10 Sensor Complementarity

No single sensor provides full state information.

Sensor	Strength	Limitation
Gyro	Fast, smooth	Drifts
Accelerometer	Gravity reference	Noisy, ambiguous
Magnetometer	Absolute heading	Disturbance sensitive
GPS	Absolute position	Low rate
Pressure	Altitude	Drift
LIDAR	Geometry	Range limits
Camera	Rich data	Computational load

This motivates sensor fusion using:

- Complementary filters,
- Extended Kalman Filters (EKF),
- Multi-sensor fusion algorithms.

11 Engineering Interpretation for Multicopters

For multicopters:

- Attitude stabilization depends primarily on IMU (gyro + accelerometer).
- Altitude hold depends on barometer and accelerometer.

- Position hold requires GPS or vision.
- Indoor flight requires optical flow or LIDAR.

Multicopters are fundamentally *estimation-driven systems*.
Control performance is only as good as the state estimate.

12 Attitude Measurement

Let F_A be the frame fixed to Earth such that the gravity vector is along the $\hat{k}_A$ axis, and the magnetic field $\vec{m}_e$ lies in the $\hat{i}_A - \hat{k}_A$ plane. The attitude of the rigid body $\mathcal{B}$ can be computed using the acceleration and the magnetic field measurements as follows. Assume that the accelerometer and the magnetometer are aligned with the frame F_D . Then,

$$a = \vec{g}|_D, \quad (20)$$

$$m = \vec{m}_e|_D. \quad (21)$$

Note that

$$\hat{k}_A|_D = \hat{g}|_D = \frac{a}{\|a\|}. \quad (22)$$

Using $\hat{k}_A \times (\alpha \hat{i}_A + \beta \hat{k}_A) = \alpha \hat{j}_A$ and the fact that $\vec{m}_e$ lies in the $\hat{i}_A - \hat{k}_A$ plane, it follows that

$$\hat{j}_A|_D = \hat{k}_A|_D \times \frac{m_c}{\|m_c\|} = \frac{a}{\|a\|} \times \frac{m_c}{\|m_c\|}. \quad (23)$$

Next,

$$\hat{i}_A|_D = \hat{k}_A|_D \times \hat{j}_A|_D = \frac{a}{\|a\|} \times \left(\frac{a}{\|a\|} \times \frac{m_c}{\|m_c\|} \right). \quad (24)$$

Finally, using the fact that

$$\mathcal{O}_{A/D} = \begin{bmatrix} \hat{i}_A|_D & \hat{j}_A|_D & \hat{k}_A|_D \end{bmatrix} \quad (25)$$

and $\mathcal{O}_{D/A} = \mathcal{O}_{A/D}^T$, it follows that

$$\mathcal{O}_{D/A} = \begin{bmatrix} \hat{i}_A|_D^T \\ \hat{j}_A|_D^T \\ \hat{k}_A|_D^T \end{bmatrix}. \quad (26)$$

13 Sensor Calibration

13.1 Magnetometer Bias Correction

The magnetic field at a point is the sum of the Earth's magnetic field $\vec{m}_e$ and the local magnetic field $\vec{m}_l$. The Earth's magnetic field $\vec{m}_e$ is fixed in the Earth-fixed frame, while the local magnetic field $\vec{m}_l$ is fixed in the magnetometer frame F_D . The magnetometer measurement m is the total magnetic field $\vec{m}_e + \vec{m}_l$ in the magnetometer frame F_D , that is,

$$m = \vec{m}_e|_D + \vec{m}_l|_D. \quad (27)$$

Note that $\vec{m}_l|_D$ is a constant, and thus is a bias. To compute $\vec{m}_l|_D$,

1. Rotate the magnetometer around each axis of F_D by 360 degrees.
2. Obtain the minimum and maximum after each rotation.
3. Average the maximum and minimum to obtain each component of $\vec{m}_l|_D$.

The corrected magnetometer reading is

$$m_c \triangleq m - \vec{m}_l|_D = \vec{m}_e|_D. \quad (28)$$

13.2 Rate-gyro Bias Correction

The 3-axis rate gyro measures

$$\omega = \vec{\omega}_{D/A}|_D + w, \quad (29)$$

where $w \in \mathbb{R}^3$ is measurement noise. The signal w is often modeled as a white noise, that is, $w \sim \mathcal{N}(b, R)$, where b is the mean and R is the variance of w . The mean b is also known as the sensor bias.

$$b = \mathbb{E}[w], \quad (30)$$

$$R = \mathbb{E}[(w - b)(w - b)^T]. \quad (31)$$

To compute b and R ,

1. Collect the rate-gyro measurements when the sensor is at rest, that is, $\vec{\omega}_{D/A}|_D = 0$.
2. The bias is approximated by $b \approx \frac{1}{N} \sum_{i=1}^N \omega_i$, where ω_i is the i th measurement.
3. The variance is approximated by $R \approx \frac{1}{N} \sum_{i=1}^N (\omega_i - b)(\omega_i - b)^T$.

13.3 GPS Variance Estimation

The GPS measurement provides

$$r = r_{cc/w}|_A + w_g, \quad (32)$$

where $w_g \in \mathbb{R}^2$ is measurement noise. The signal w_g is often modeled as a white noise, that is, $w \sim \mathcal{N}(b, R)$, where b is the mean and R is the variance of w . The mean b is also known as the sensor bias.

$$b = \mathbb{E}[w], \quad (33)$$

$$R = \mathbb{E}[(w - b)(w - b)^T]. \quad (34)$$

To compute b and R ,

- 1.

13.4 Sensor Alignment

Let F_A be the frame fixed to Earth such that the gravity vector is along the $\hat{k}_A$ axis, and the magnetic field $\vec{m}_e$ lies in the $\hat{i}_A - \hat{k}_A$ plane. Assume that the sensor frame F_D is aligned with F_A . Then,

$$\hat{k}_D = \hat{k}_A = \hat{g} \triangleq \frac{\vec{g}}{g}. \quad (35)$$

14 Accelerometer Calibration

14.1 Measurement Model

A tri-axial accelerometer measurement is modeled as

$$a_m = K a_t + b + n, \quad (36)$$

where $a_m \in \mathbb{R}^3$ is the measured acceleration, $a_t \in \mathbb{R}^3$ is the true specific force, $b \in \mathbb{R}^3$ is the accelerometer bias, $K \in \mathbb{R}^{3 \times 3}$ is the scale factor matrix, and n is measurement noise. For the BNO055, K is assumed to satisfy

$$K = \text{diag}(k_x, k_y, k_z). \quad (37)$$

14.2 Static Calibration Assumption

During calibration, the sensor is held stationary, implying

$$\|a_t\| = g, \quad (38)$$

where $g = 9.80665 \text{ m/s}^2$ is gravitational acceleration. By placing the sensor in six static orientations, the true acceleration satisfies

$$a_t \in \{(\pm g, 0, 0), (0, \pm g, 0), (0, 0, \pm g)\}. \quad (39)$$

14.3 Six-Pose Measurements

Let the averaged accelerometer measurements be denoted by

$$a_x^+, a_x^- \quad \text{for } +x \text{ and } -x \text{ orientations,} \quad (40)$$

$$a_y^+, a_y^- \quad \text{for } +y \text{ and } -y \text{ orientations,} \quad (41)$$

$$a_z^+, a_z^- \quad \text{for } +z \text{ and } -z \text{ orientations.} \quad (42)$$

14.4 Bias Estimation

The accelerometer bias is given by

$$b_x = \frac{a_x^+ + a_x^-}{2}, \quad b_y = \frac{a_y^+ + a_y^-}{2}, \quad b_z = \frac{a_z^+ + a_z^-}{2}. \quad (43)$$

14.5 Scale Factor Estimation

The per-axis scale factors are given by

$$k_x = \frac{a_x^+ - a_x^-}{2g}, \quad k_y = \frac{a_y^+ - a_y^-}{2g}, \quad k_z = \frac{a_z^+ - a_z^-}{2g}. \quad (44)$$

14.6 Calibrated Acceleration

Given a raw accelerometer measurement a_m , the calibrated acceleration is

$$a_c = K^{-1} (a_m - b). \quad (45)$$

After calibration, a stationary sensor satisfies $\|a_c\| \approx g$.

14.7 Noise Covariance Estimation

For stationary calibrated measurements $a_c(k)$, the accelerometer noise covariance is estimated as

$$\Sigma_a = \mathbb{E} \left[(a_c - \bar{a}_c)(a_c - \bar{a}_c)^\top \right], \quad (46)$$

which is often approximated as diagonal,

$$\Sigma_a \approx \text{diag}(\sigma_x^2, \sigma_y^2, \sigma_z^2). \quad (47)$$

15 Gyroscope Calibration

15.1 Measurement Model

A tri-axial gyroscope measurement is modeled as

$$\omega_m = K_\omega \omega_t + b_\omega + n_\omega, \quad (48)$$

where $\omega_m \in \mathbb{R}^3$ is the measured angular velocity, $\omega_t \in \mathbb{R}^3$ is the true angular velocity, $b_\omega \in \mathbb{R}^3$ is the gyroscope bias, $K_\omega \in \mathbb{R}^{3 \times 3}$ is the scale factor matrix, and n_ω is measurement noise. For the BNO055 and similar MEMS gyroscopes, scale factor is assumed to be of the form

$$K_\omega = \text{diag}(k_{\omega_x}, k_{\omega_y}, k_{\omega_z}). \quad (49)$$

15.2 Static Calibration Assumption

During calibration, the sensor is held stationary, implying

$$\omega_t = 0. \quad (50)$$

Under this condition, the measurement model reduces to

$$\omega_m = b_\omega + n_\omega. \quad (51)$$

15.3 Bias Estimation

The gyroscope bias is estimated by time-averaging the measured angular velocity while the sensor is stationary, that is,

$$b_\omega = \frac{1}{N} \sum_{k=1}^N \omega_m(k). \quad (52)$$

This estimator is unbiased under the assumption of zero-mean noise.

15.4 Bias-Corrected Measurement

After bias removal, the calibrated gyroscope measurement is

$$\omega_c = \omega_m - b_\omega. \quad (53)$$

15.5 Noise Covariance Estimation

The gyroscope noise covariance is estimated from stationary, bias-corrected measurements:

$$\Sigma_\omega = \mathbb{E} \left[(\omega_c - \bar{\omega}_c)(\omega_c - \bar{\omega}_c)^\top \right], \quad (54)$$

where $\bar{\omega}_c$ denotes the sample mean. In practice, a diagonal approximation is used, that is,

$$\Sigma_\omega \approx \text{diag}(\sigma_{\omega_x}^2, \sigma_{\omega_y}^2, \sigma_{\omega_z}^2). \quad (55)$$

16 Magnetometer Calibration

16.1 Measurement Model

A tri-axial magnetometer measurement is modeled as

$$m_m = S R m_t + b_m + n_m, \quad (56)$$

where $m_m \in \mathbb{R}^3$ is the measured magnetic field, $m_t \in \mathbb{R}^3$ is the true magnetic field expressed in the sensor frame, $b_m \in \mathbb{R}^3$ is the hard-iron bias, $S \in \mathbb{R}^{3 \times 3}$ is the soft-iron distortion matrix, $R \in SO(3)$ is a misalignment matrix, and n_m is measurement noise. In practice, the soft-iron and misalignment effects are combined into a single full-rank matrix

$$K_m \triangleq SR, \quad (57)$$

yielding

$$m_m = K_m m_t + b_m + n_m. \quad (58)$$

16.2 Calibration Objective

In a magnetically homogeneous environment, the magnitude of the true magnetic field is constant, that is,

$$\|m_t\| = M, \quad (59)$$

where $M > 0$ is the local geomagnetic field strength.

The objective of magnetometer calibration is to determine K_m and b_m such that the calibrated measurements satisfy

$$\|m_c\| = M, \quad m_c = K_m^{-1}(m_m - b_m). \quad (60)$$

16.3 Ellipsoidal Measurement Model

Due to hard-iron and soft-iron effects, the raw magnetometer measurements satisfy the quadratic form

$$(m_m - b_m)^\top W (m_m - b_m) = M^2, \quad (61)$$

where $W \in \mathbb{R}^{3 \times 3}$ is a symmetric positive-definite matrix. This equation represents an ellipsoid in measurement space.

16.4 Ellipsoid Fitting

Given N magnetometer samples $\{m_m(k)\}_{k=1}^N$ collected over a wide range of orientations, the parameters (W, b_m) are obtained by solving a constrained least-squares problem:

$$\min_{W, b_m} \sum_{k=1}^N \left[(m_m(k) - b_m)^\top W (m_m(k) - b_m) - M^2 \right]^2, \quad (62)$$

subject to $W \succ 0$.

16.5 Recovery of Calibration Matrix

Let the eigenvalue decomposition of W be

$$W = Q\Lambda Q^\top, \quad (63)$$

where Q is orthogonal and Λ is diagonal with positive entries.

A valid calibration matrix is obtained as

$$K_m = Q \Lambda^{1/2}. \quad (64)$$

The calibrated magnetometer measurement is then

$$m_c = K_m^{-1}(m_m - b_m). \quad (65)$$

16.6 Noise Covariance Estimation

After calibration, the magnetometer noise covariance is estimated from stationary data:

$$\Sigma_m = \mathbb{E} \left[(m_c - \bar{m}_c)(m_c - \bar{m}_c)^\top \right]. \quad (66)$$

A diagonal approximation is often used in practice:

$$\Sigma_m \approx \text{diag}(\sigma_{mx}^2, \sigma_{my}^2, \sigma_{mz}^2). \quad (67)$$

16.7 Remarks

- Hard-iron effects produce constant measurement offsets.
- Soft-iron effects distort the measurement space into an ellipsoid.
- Accurate calibration requires excitation over a wide range of orientations.
- The resulting calibration parameters are suitable for use in attitude estimation and sensor fusion algorithms.

Sensor	Measurement Relation $y = h(x)$	Comment / Remark
Accelerometer	$y_{acc} = R(q)^T(\dot{v} - g_I)$	Measures specific force, not direct inertial acceleration. Depends on attitude and acceleration from process model; not purely state-only unless augmented.
Gyroscope	$y_{gyro} = \omega$	Direct measurement of body angular velocity. Primary sensor for attitude propagation.
Magnetometer	$y_{mag} = R(q)^T m_I$	Measures Earth magnetic field in body frame. Used mainly for yaw/heading correction.
Barometer	$y_{baro} = p_z$	Provides altitude estimate from atmospheric pressure. Typically noisy and slowly drifting.
Pitot Tube	$y_{pitot} = e_1^T R(q)^T (v - v_{wind})$ or $y_{pitot} = \ v - v_{wind}\ $	Measures air-relative speed, not ground speed. Depends on unknown wind if not estimated separately.
GPS / GNSS	$y_{gps} = \begin{bmatrix} p \\ v \end{bmatrix}$	Direct global position and velocity measurement. Low rate, subject to signal blockage and noise.
RTK GPS	$y_{rtk} = \begin{bmatrix} p \\ v \end{bmatrix}$	Same as GPS but much higher precision (centimeter-level position).
ToF Sensor	$y_{tof} \approx p_z$	Measures slant range to nearby surface; often approximated as altitude for near-level flight.
LiDAR	$y_{lidar} = h(p, q, \mathcal{M})$	Range depends on position, attitude, and environment map. Used for localization and mapping.
Ultrasonic Sensor	$y_{ultra} \approx p_z$	Short-range altitude sensing. Limited range and affected by surface properties.
Radar Altimeter	$y_{radar} \approx p_z$	Reliable altitude above ground, especially outdoors. Less sensitive to lighting conditions.
Camera (Monocular / Stereo / Depth)	$y_{cam,i} = \pi(R(q)^T (P_i - p))$	Feature-based geometric measurement. Depends on position and attitude relative to landmarks.
Optical Flow Sensor	$y_{flow} \approx \frac{1}{p_z} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} R(q)^T v$	Indirect horizontal velocity sensing. Scaled by altitude and affected by attitude.
Motion Capture System	$y_{mocap} = \begin{bmatrix} p \\ q \end{bmatrix}$	High-precision external pose measurement. Common in indoor labs.
Lighthouse System	$y_{lh} = \begin{bmatrix} p \\ q \end{bmatrix}$	External localization system similar to motion capture but lower cost.
3DoF Rotating Platform	$y_{plat} = q$	Used mainly for testing/calibration of attitude estimation.

Table 1: Measurement models for common UAV sensors in the nonlinear form $y = h(x)$, with state $x = [p, v, q, \omega]^T$. Page 16 of 16